free_choice_2

March 23, 2026

1 Free Choice 2

2 Imports and Data Loading

Imports

[1]:
```
# imports
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit
```

Data Loading

[2]:
```
exercise_1_data = np.loadtxt("../data/session1/ex1_trial_2/theta.txt",␣
↪delimiter='\t', comments='#', unpack=True, skiprows=2) # tab separated
exercise_2_data_180 = np.loadtxt("../data/session1/ex2_trial_2/180/theta.txt",␣
↪delimiter='\t', comments='#', unpack=True, skiprows=2) # tab separated
exercise_2_data_360 = np.loadtxt("../data/session1/ex2_trial_2/360/theta.txt",␣
↪delimiter='\t', comments='#', unpack=True, skiprows=2) # tab separated

exercise_1_cossq_data = np.loadtxt("../data/session1/ex1_trial_2/
↪cosinesquaredtheta.txt", delimiter='\t', comments='#', unpack=True,␣
↪skiprows=2) # tab separated

# Malus' Law

# exercise 1

ex1_position = exercise_1_data[0] # Sensor Position (Radians)
ex1_intensity = exercise_1_data[1] # Light Intensity (Volts)

# exercise 2

ex2_position_180 = exercise_2_data_180[0] # Sensor Position (Radians)
ex2_intensity_180 = exercise_2_data_180[1] # Light Intensity (Volts)

ex2_position_360 = exercise_2_data_360[0] # Sensor Position (Radians)
ex2_intensity_360 = exercise_2_data_360[1] # Light Intensity (Volts)
```

```
ex2_position_combined = np.concatenate((ex2_position_180, ex2_position_360 + np.
 ↪pi), axis=0)
ex2_intensity_combined = np.concatenate((ex2_intensity_180, ex2_intensity_360),␣
 ↪axis=0)

# Brewster

# exercise 3

exercise_3_data_no_polarizer = np.loadtxt("../data/clean_data/no_polarizer.
 ↪txt", delimiter='\t', comments='#', unpack=True, skiprows=2) # tab separated
no_polarizer_position = exercise_3_data_no_polarizer[0] - 85 # actually starts␣
 ↪at 95. This is wrt the rotational measuring thing
no_polarizer_intensity = exercise_3_data_no_polarizer[1]

exercise_3_data_horizontal = np.loadtxt("../data/clean_data/horizontal.txt",␣
 ↪delimiter='\t', comments='#', unpack=True, skiprows=2) # tab separated
horizontal_position =  exercise_3_data_horizontal[0] - 85 # actually starts at␣
 ↪95
horizontal_intensity = exercise_3_data_horizontal[1]

exercise_3_data_vertical = np.loadtxt("../data/clean_data/vertical.txt",␣
 ↪delimiter='\t', comments='#', unpack=True, skiprows=2) # tab separated
vertical_position = exercise_3_data_vertical[0] - 85 # actually starts at 95
vertical_intensity = exercise_3_data_vertical[1]
```

3 Uncertainties

[3]:
```
# there is a big assumption that the uncertainty of the sensor does not depend␣
 ↪on intensity.

malus_1_uncertainty = np.loadtxt("../data/session3/unc/malus1.txt",␣
 ↪delimiter='\t', comments='#', unpack=True, skiprows=2) # tab separated
brewster_1_uncertainty = np.loadtxt("../data/session3/unc/brewster1.txt",␣
 ↪delimiter='\t', comments='#', unpack=True, skiprows=2) # tab separated
brewster_10_uncertainty = np.loadtxt("../data/session3/unc/brewster10.txt",␣
 ↪delimiter='\t', comments='#', unpack=True, skiprows=2) # tab separated
```

[4]:
```
print(f"malus sensor 1 gain:\nmean: {np.
 ↪mean(malus_1_uncertainty[1])}\nuncertainty: {np.
 ↪std(malus_1_uncertainty[1])}")
print(f"brewster sensor 1 gain:\nmean: {np.
 ↪mean(brewster_1_uncertainty[1])}\nuncertainty: {np.
 ↪std(brewster_1_uncertainty[1])}")
```

```
print(f"brewster sensor 10 gain:\nmean: {np.
↪mean(brewster_10_uncertainty[1])}\nuncertainty: {np.
↪std(brewster_10_uncertainty[1])}")
```

```
malus sensor 1 gain:
mean: 1.2314533604777416
uncertainty: 0.010426601916429589
brewster sensor 1 gain:
mean: 1.218340076775432
uncertainty: 0.0033007650473998004
brewster sensor 10 gain:
mean: 1.2373230562827224
uncertainty: 0.002754706969711812
```

4 Model Functions

[5]:
```
# function for calculating chi squared reduced.
def chi_squared_reduced(y_measure, y_predict, errors, number_of_parameters):
    """Calculates the reduced chi squared of data, prediction and errors."""
    return (np.sum(((y_measure - y_predict) ** 2 )/ (errors ** 2)))/(y_measure.
↪size - number_of_parameters)
```

4.1 Malus

[6]:
```
# fit raw data to cos^2 theta

def cossqtheta(theta, a, b, c):
    return a * (np.cos(b * (theta - c)) ** 2)
```

[7]:
```
def coscos(theta, a, b, c, d):
    return a * (np.cos(b * (theta - c)) ** 2) * (np.cos(b * ((d - theta) - c))␣
↪** 2)
```

4.2 Brewster

[8]:
```
# Snell and Reflectance equations combined.

def perpendicular_reflectance(theta_in, n_2, a, b, c):
    theta_i = theta_in + b # shift
    n_1 = 1 # by assumptions in lab manual
    theta_f = np.arcsin((n_1/n_2) * np.sin(theta_i))

    # formula to curve fit.
    return (((n_1 * theta_i - (n_2 * np.cos(theta_f))) / (n_1 * theta_i + (n_2␣
↪* np.cos(theta_f)))) ** 2) * c + a  # amplitude is c, shift is a
```

[9]:
```
def parallel_reflectance(theta_in, n_2, a, b, c):
    theta_i = theta_in + b # shift
    n_1 = 1 # by assumptions in lab manual
    theta_f = np.arcsin((n_1/n_2) * np.sin(theta_i))

    # formula to curve fit.
    return (((n_1 * np.cos(theta_f) - (n_2 * theta_i)) / (n_1 * np.cos(theta_f)␣
↪+ (n_2 * theta_i))) ** 2) * c + a # amplitude is c, shift is a
```

[10]:
```
def all_reflectance(theta_in, n_2, a, b, c):
    return (parallel_reflectance(theta_in, n_2, a, b, c) +␣
↪perpendicular_reflectance(theta_in, n_2, a, b, c)) / 2
```

5 Curve Fit and Model Parameters

5.1 Malus

[11]:
```
# ex1 2 polarizers
popt, pcov = curve_fit(cossqtheta, xdata=ex1_position, ydata=ex1_intensity,␣
↪absolute_sigma=True, sigma=0.01) #, absolute_sigma=True, sigma =␣
↪propTimeError)
pstd = np.sqrt(np.diag(pcov))

# ex2 3 polarizers
popt2, pcov2 = curve_fit(coscos, xdata=ex2_position_combined,␣
↪ydata=ex2_intensity_combined, absolute_sigma=True, sigma=0.01)
pstd2 = np.sqrt(np.diag(pcov2))

# ex1
print("chi squared reduced of 2 polarizers␣
↪=",chi_squared_reduced(ex1_intensity, cossqtheta(ex1_position, popt[0],␣
↪popt[1], popt[2]), 1, len(popt)))
print(f"Value of a: {popt[0]} \nValue of b: {popt[1]} \nValue of c: {popt[2]}")
print(f"Uncertainty in a: {pstd[0]} \nUncertainty in b: {pstd[1]} \nUncertainty␣
↪in c: {pstd[2]}")
print("_____________________________________________________________")
# ex2
print("chi squared reduced of 3 polarizers␣
↪=",chi_squared_reduced(ex2_position_combined, coscos(ex2_intensity_combined,␣
↪popt2[0], popt2[1], popt2[2], popt2[3]), 1, len(popt2)))
print(f"Value of a: {popt2[0]} \nValue of b: {popt2[1]} \nValue of c:␣
↪{popt2[2]} \nValue of d: {popt2[3]}")
print(f"Uncertainty in a: {pstd2[0]} \nUncertainty in b: {pstd2[1]}␣
↪\nUncertainty in c: {pstd2[2]} \nUncertainty in d: {pstd2[3]}")
```

```
chi squared reduced of 2 polarizers = 0.0016490399732635114
Value of a: 4.499491008289387
```

```
Value of b: 1.0061932109053942
Value of c: 0.5753631935406328
Uncertainty in a: 0.0006219366691890683
Uncertainty in b: 0.00012001051358845225
Uncertainty in c: 0.00015626329197983388
-----------------------------------------------------------
chi squared reduced of 3 polarizers = 9.890910821959395
Value of a: 5.0746276846373375
Value of b: 0.9925153046850576
Value of c: 0.5273667685931266
Value of d: -0.2507116881884684
Uncertainty in a: 0.0029608314948976837
Uncertainty in b: 0.00010934463392909071
Uncertainty in c: 0.00042683652302864924
Uncertainty in d: 0.0008366070083360444
```

5.2 Brewster

[12]:
```
# curve fit horizontal
popt_h, pcov_h = curve_fit(perpendicular_reflectance,␣
 ↪xdata=horizontal_position, ydata=horizontal_intensity, absolute_sigma=True,␣
 ↪sigma=0.003, maxfev=10000)
pstd_h = np.sqrt(np.diag(pcov_h))

# vertical
popt_v, pcov_v = curve_fit(parallel_reflectance, xdata=vertical_position,␣
 ↪ydata=vertical_intensity, absolute_sigma=True, sigma=0.003)
pstd_v = np.sqrt(np.diag(pcov_v))

# none
popt_n, pcov_n = curve_fit(all_reflectance, xdata=no_polarizer_position,␣
 ↪ydata=no_polarizer_intensity, absolute_sigma=True, sigma=0.003)
pstd_n = np.sqrt(np.diag(pcov_n))


print("chi squared reduced of horizontal␣
 ↪=",chi_squared_reduced(horizontal_intensity,␣
 ↪perpendicular_reflectance(horizontal_position, popt_h[0], popt_h[1],␣
 ↪popt_h[2], popt_h[3]), 1, len(popt_h)))
print(f"Value of n_2: {popt_h[0]} \nValue of a: {popt_h[1]} \nValue of b:␣
 ↪{popt_h[2]} \nValue of c: {popt_h[3]}")
print(f"Uncertainty in n_2: {pstd_h[0]} \nUncertainty in a: {pstd_h[1]}␣
 ↪\nUncertainty in b: {pstd_h[2]} \nUncertainty in c: {pstd_h[3]}")
print("_____________________________________________________________")
```

```
print("chi squared reduced of vertical␣
↪=",chi_squared_reduced(vertical_intensity,␣
↪parallel_reflectance(vertical_position, popt_v[0], popt_v[1], popt_v[2],␣
↪popt_v[3]), 1, len(popt_v)))
print(f"Value of n_2: {popt_v[0]} \nValue of a: {popt_v[1]} \nValue of b:␣
↪{popt_v[2]} \nValue of c: {popt_v[3]}")
print(f"Uncertainty in n_2: {pstd_v[0]} \nUncertainty in a: {pstd_v[1]}␣
↪\nUncertainty in b: {pstd_v[2]} \nUncertainty in c: {pstd_v[3]}")
print("_____________________________________________________________")
print("chi squared reduced of none␣
↪=",chi_squared_reduced(no_polarizer_intensity,␣
↪all_reflectance(no_polarizer_position, popt_n[0], popt_n[1], popt_n[2],␣
↪popt_n[3]), 1, len(popt_n)))
print(f"Value of n_2: {popt_n[0]} \nValue of a: {popt_n[1]} \nValue of b:␣
↪{popt_n[2]} \nValue of c: {popt_n[3]}")
print(f"Uncertainty in n_2: {pstd_n[0]} \nUncertainty in a: {pstd_n[1]}␣
↪\nUncertainty in b: {pstd_n[2]} \nUncertainty in c: {pstd_n[3]}")
```

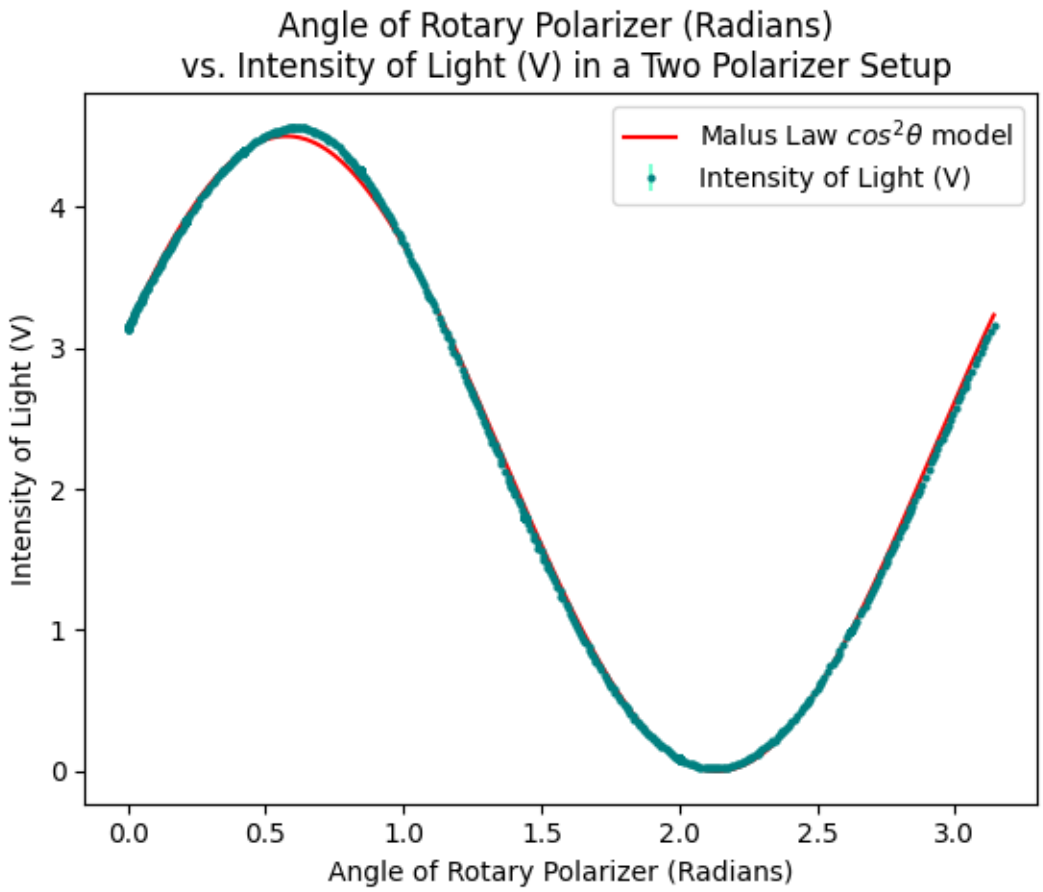
```
chi squared reduced of horizontal = 0.00536869864070037
Value of n_2: 31.683837929897763
Value of a: 0.07001390817481476
Value of b: -36.558699141489235
Value of c: 37.54100340904807
Uncertainty in n_2: 0.012598295504611097
Uncertainty in a: 0.00018314204610450627
Uncertainty in b: 0.011256274585139352
Uncertainty in c: 0.026169384042817764
_____________________________________________________________
chi squared reduced of vertical = 0.4251171827693606
Value of n_2: 0.9999986943541054
Value of a: 0.49113434539786294
Value of b: 1.612974958360588
Value of c: 0.4122284776167357
Uncertainty in n_2: 1.4139533194646505e-05
Uncertainty in a: 0.0015748616962502641
Uncertainty in b: 0.002235090379890384
Uncertainty in c: 0.0016698847783698663
_____________________________________________________________
chi squared reduced of none = 0.8137598605795068
Value of n_2: 0.9999924697628563
Value of a: 0.6748943004819753
Value of b: 0.465686331820694
Value of c: 0.6240877008680114
Uncertainty in n_2: 1.060962156154333e-05
Uncertainty in a: 0.0018135107344042503
Uncertainty in b: 0.0017000893576645298
Uncertainty in c: 0.0019200666748138172
```

```
/var/folders/b4/87nbcb5j6d772d537gj3mnc80000gn/T/ipykernel_72228/1316014866.py:4
: RuntimeWarning: invalid value encountered in arcsin
  theta_f = np.arcsin((n_1/n_2) * np.sin(theta_i))
/var/folders/b4/87nbcb5j6d772d537gj3mnc80000gn/T/ipykernel_72228/1250819704.py:6
: RuntimeWarning: invalid value encountered in arcsin
  theta_f = np.arcsin((n_1/n_2) * np.sin(theta_i))
```

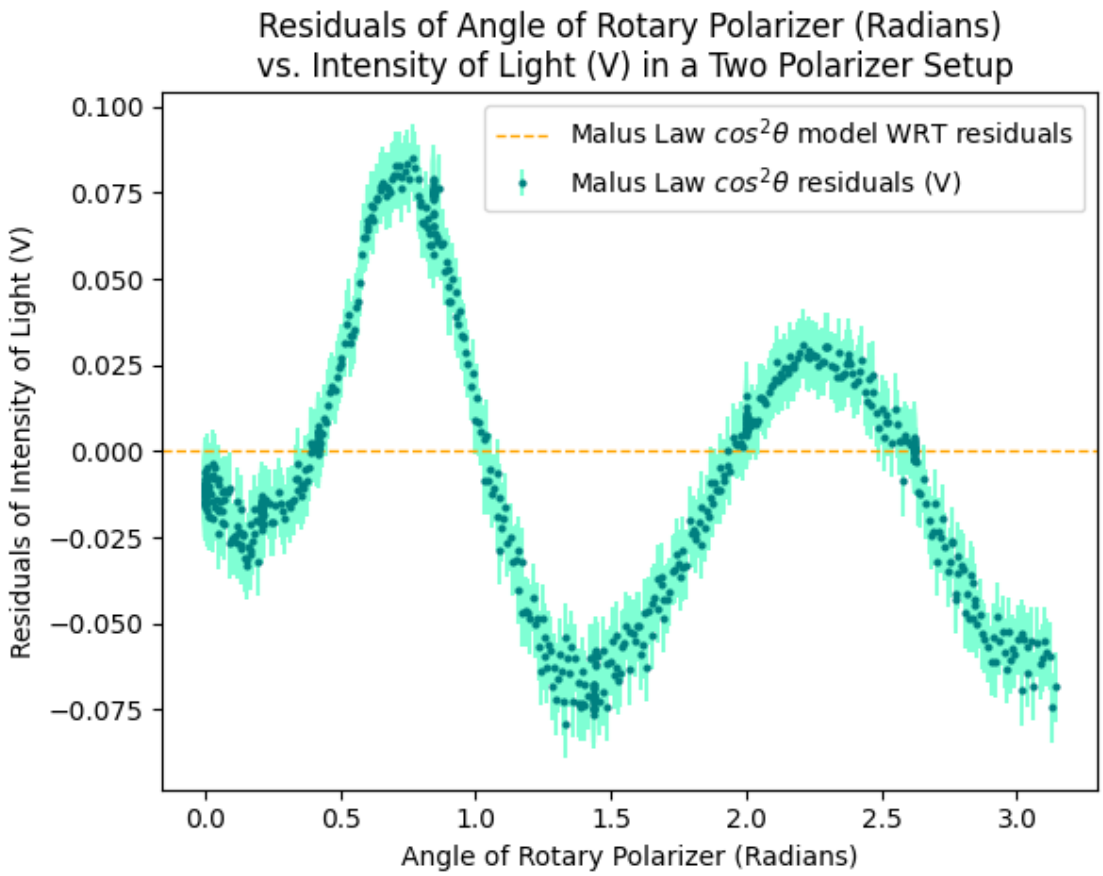
6 Graphs for Curve Fit

6.1 Malus

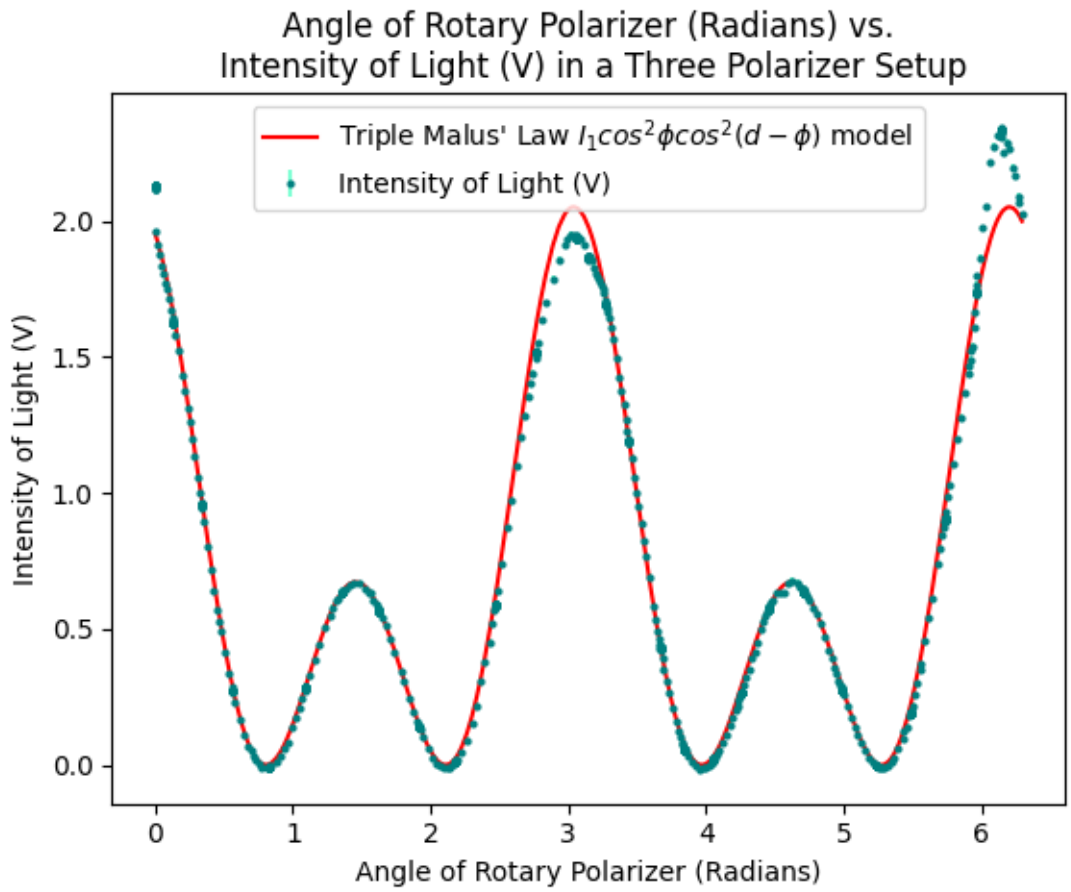
[13]:
```
# scatter error bar
plt.errorbar(x = ex1_position, y = ex1_intensity, yerr=0.01, fmt= 'o',
            ms=2, color='teal', ecolor='aquamarine', label="Intensity of Light␣
 ↪(V)")

# curve fit plot
plt.plot(ex1_position, cossqtheta(ex1_position, popt[0], popt[1], popt[2]),␣
 ↪label = "Malus Law $cos^2\\theta$ model", color='red')

# plt.plot(ex1_position, np.cos(ex1_position - 0.58) ** 2 * 4.5, label = "Malus␣
 ↪Law $cos^2\\theta$ model", color='blue') # actual malus law

# Legend and titles
plt.legend()
plt.xlabel("Angle of Rotary Polarizer (Radians)")
plt.ylabel("Intensity of Light (V)")
plt.title("Angle of Rotary Polarizer (Radians) \n vs. Intensity of Light (V) in␣
 ↪a Two Polarizer Setup")

plt.show()
```



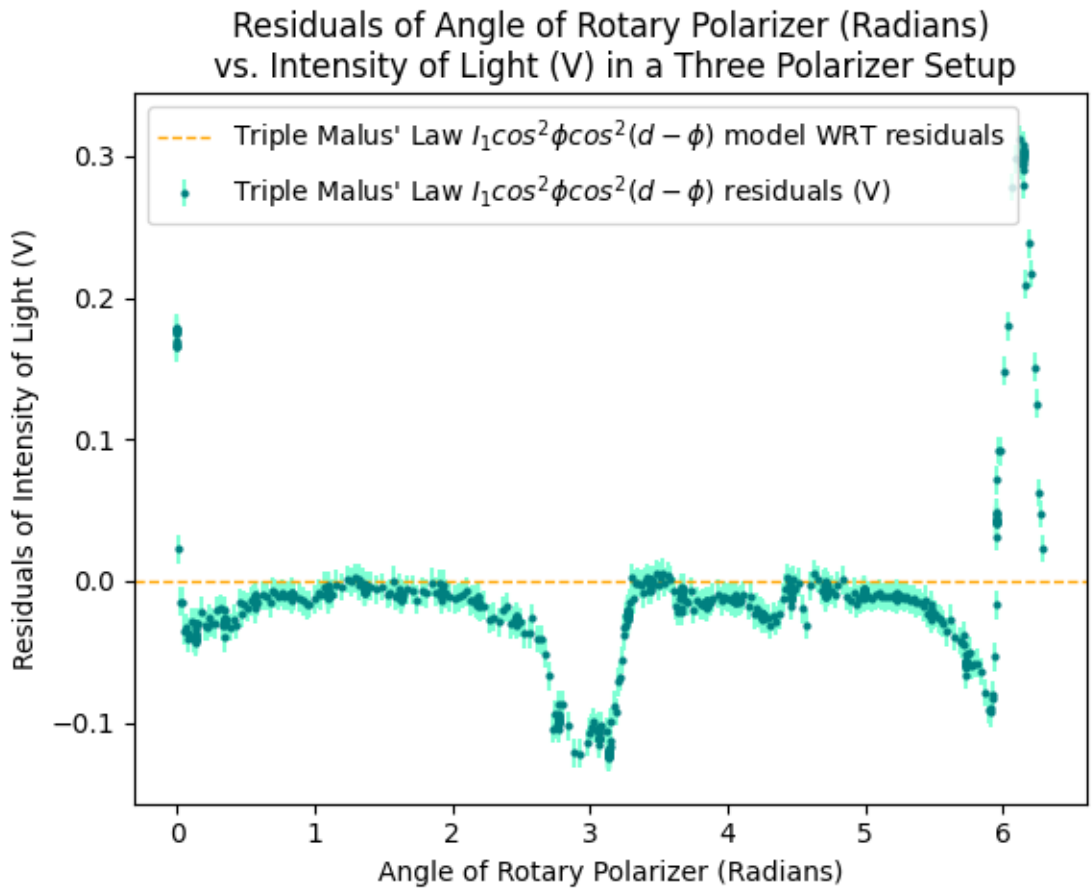
[14]:
```
residuals = ex1_intensity - cossqtheta(ex1_position, popt[0], popt[1], popt[2])

plt.errorbar(x = ex1_position, y = residuals, yerr = 0.01, fmt= 'o', ms=2,␣
↪label="Malus Law $cos^2\\theta$ residuals (V)",ls='', color='teal',␣
↪ecolor='aquamarine')
plt.axhline(y=0, color='orange', linestyle='--', linewidth=1, label="Malus Law␣
↪$cos^2\\theta$ model WRT residuals")

# legend and titles
plt.legend()
plt.xlabel("Angle of Rotary Polarizer (Radians)")
plt.ylabel("Residuals of Intensity of Light (V)")
plt.title("Residuals of Angle of Rotary Polarizer (Radians)\n vs. Intensity of␣
↪Light (V) in a Two Polarizer Setup")

plt.show()
```



[15]:
```
# scatter error bar
plt.errorbar(x = ex2_position_combined, y = ex2_intensity_combined, yerr=0.01,␣
↪fmt= 'o',
             ms=2, color='teal', ecolor='aquamarine', label="Intensity of Light␣
↪(V)")

# curve fit plot
plt.plot(ex2_position_combined, coscos(ex2_position_combined, popt2[0],␣
↪popt2[1], popt2[2], popt2[3]), label = "Triple Malus' Law $I_1cos^2\\phi␣
↪cos^2(d - \phi)$ model", color='red')

# Legend and titles
plt.legend()
plt.xlabel("Angle of Rotary Polarizer (Radians)")
plt.ylabel("Intensity of Light (V)")
plt.title("Angle of Rotary Polarizer (Radians) vs.\n Intensity of Light (V) in␣
↪a Three Polarizer Setup")

plt.show()
```



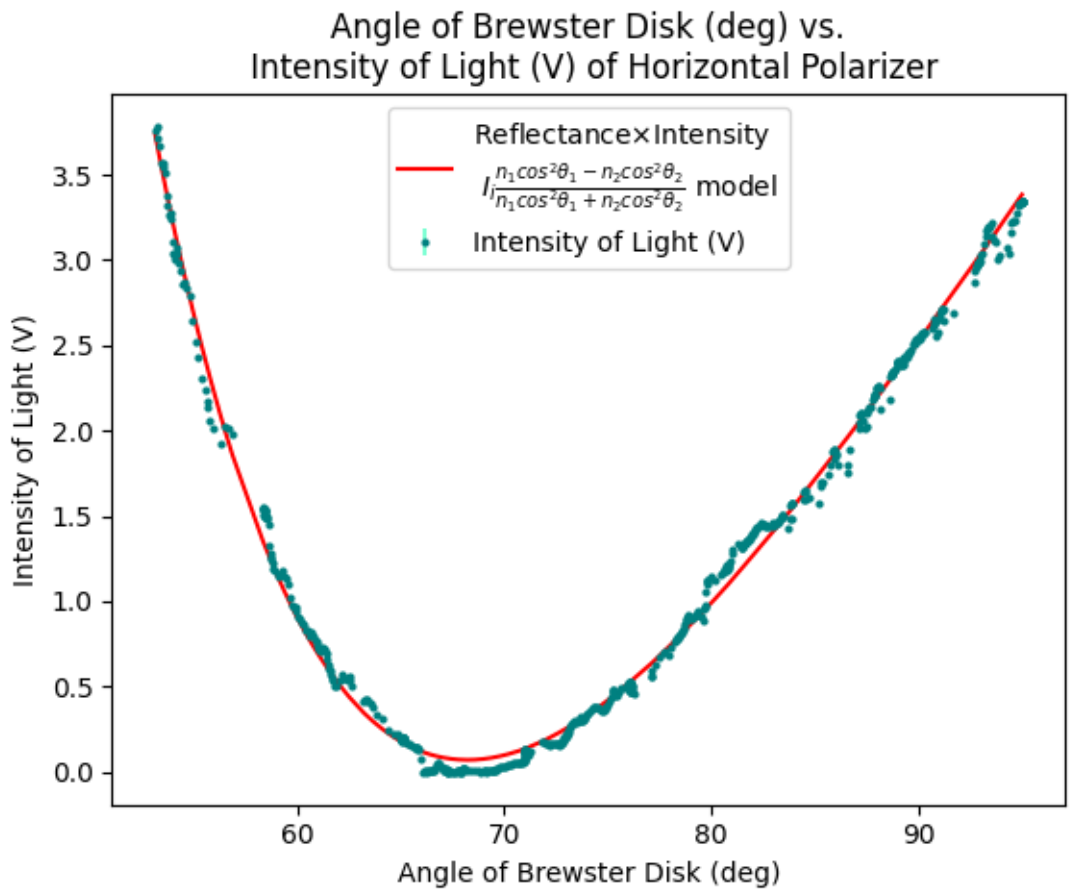
[16]:
```
residuals = ex2_intensity_combined - coscos(ex2_position_combined, popt2[0],␣
↪popt2[1], popt2[2], popt2[3])

plt.errorbar(x = ex2_position_combined, y = residuals, yerr = 0.01, fmt= 'o',␣
↪ms=2, label="Triple Malus' Law $I_1cos^2\\phi cos^2(d - \phi)$ residuals␣
↪(V)",ls='', color='teal', ecolor='aquamarine')
plt.axhline(y=0, color='orange', linestyle='--', linewidth=1, label="Triple␣
↪Malus' Law $I_1cos^2\\phi cos^2(d - \phi)$ model WRT residuals")

# legend and titles
plt.legend()
plt.xlabel("Angle of Rotary Polarizer (Radians)")
plt.ylabel("Residuals of Intensity of Light (V)")
plt.title("Residuals of Angle of Rotary Polarizer (Radians)\n vs. Intensity of␣
↪Light (V) in a Three Polarizer Setup")

plt.show() # notice the spike at the end
```



6.2 Brewster

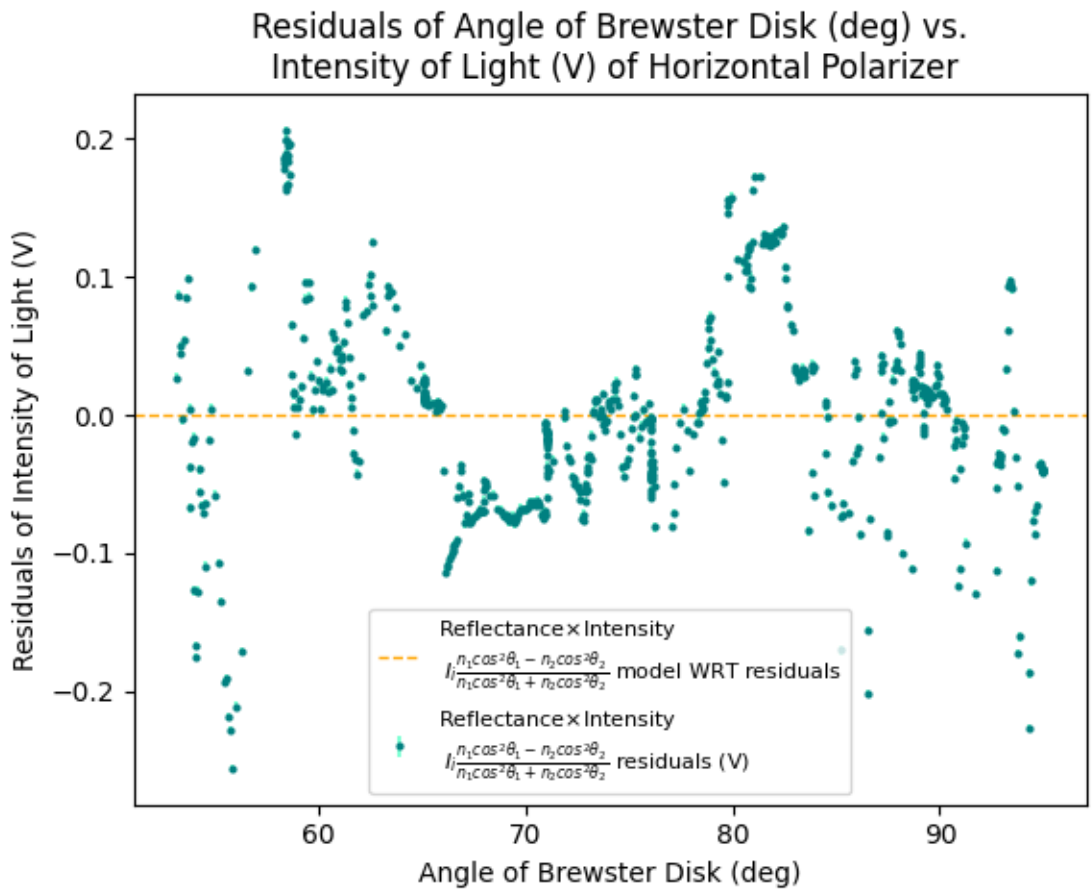
[17]:
```
# scatter error bar
plt.errorbar(x = horizontal_position, y = horizontal_intensity, yerr = 0.003,␣
↪fmt= 'o',
            ms=2, color='teal', ecolor='aquamarine', label="Intensity of Light␣
↪(V)")

plt.plot(horizontal_position, perpendicular_reflectance(horizontal_position,␣
↪popt_h[0], popt_h[1], popt_h[2], popt_h[3]), label =␣
↪"Reflectance$\\times$Intensity \n $I_i\\frac{n_1cos^2\\theta_1 -␣
↪n_2cos^2\\theta_2}{n_1cos^2\\theta_1 + n_2cos^2\\theta_2}$ model",␣
↪color='red')


# Legend and titles
plt.legend()
plt.xlabel("Angle of Brewster Disk (deg)")
```

```
plt.ylabel("Intensity of Light (V)")
plt.title("Angle of Brewster Disk (deg) vs.\n Intensity of Light (V) of␣
↪Horizontal Polarizer")

plt.show()
```



[18]:
```
residuals = horizontal_intensity -␣
↪perpendicular_reflectance(horizontal_position, popt_h[0], popt_h[1],␣
↪popt_h[2], popt_h[3])

plt.errorbar(x = horizontal_position, y = residuals, yerr = 0.003, fmt= 'o',␣
↪ms=2, label="Reflectance$\\times$Intensity \n $I_i\\frac{n_1cos^2\\theta_1 -␣
↪n_2cos^2\\theta_2}{n_1cos^2\\theta_1 + n_2cos^2\\theta_2}$ residuals␣
↪(V)",ls='', color='teal', ecolor='aquamarine')
plt.axhline(y=0, color='orange', linestyle='--', linewidth=1,␣
↪label="Reflectance$\\times$Intensity \n $I_i\\frac{n_1cos^2\\theta_1 -␣
↪n_2cos^2\\theta_2}{n_1cos^2\\theta_1 + n_2cos^2\\theta_2}$ model WRT␣
↪residuals")
```

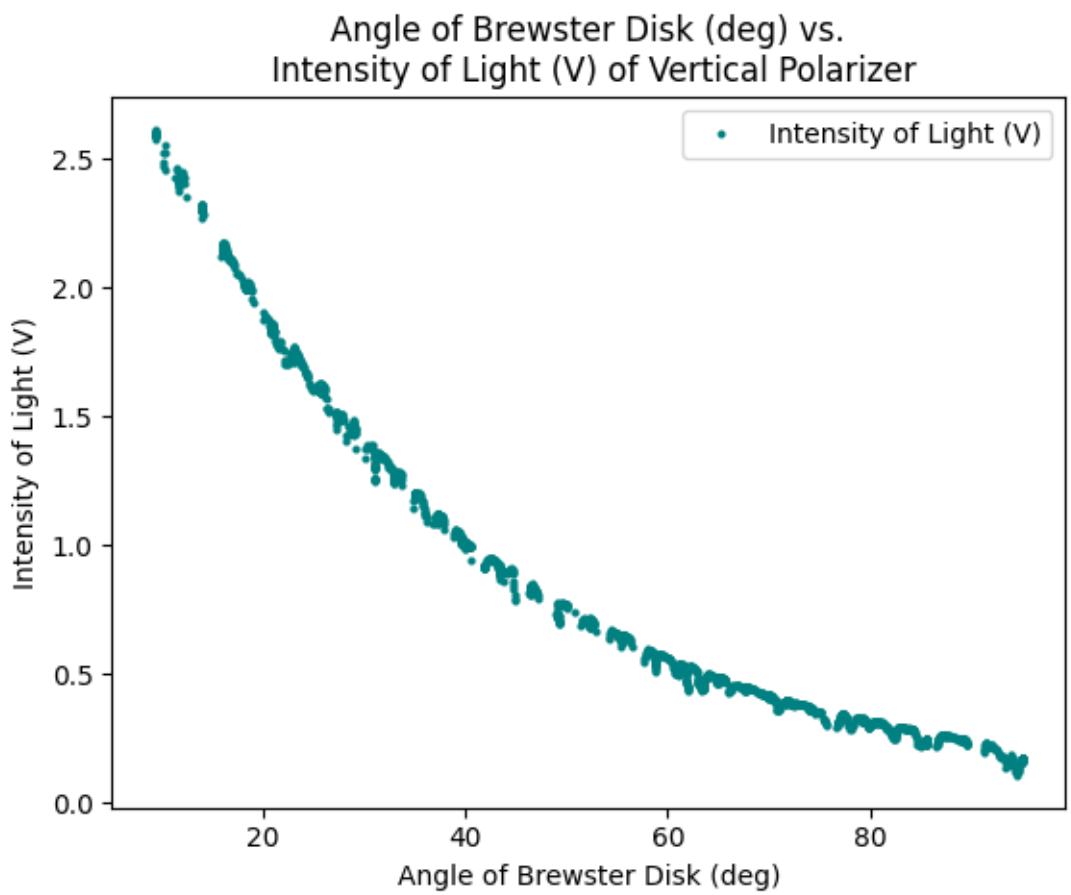
```
# legend and titles
plt.legend(fontsize=8)
plt.xlabel("Angle of Brewster Disk (deg)")
plt.ylabel("Residuals of Intensity of Light (V)")
plt.title("Residuals of Angle of Brewster Disk (deg) vs.\n Intensity of Light␣
↪(V) of Horizontal Polarizer")

plt.show() # notice the spike at the end
```

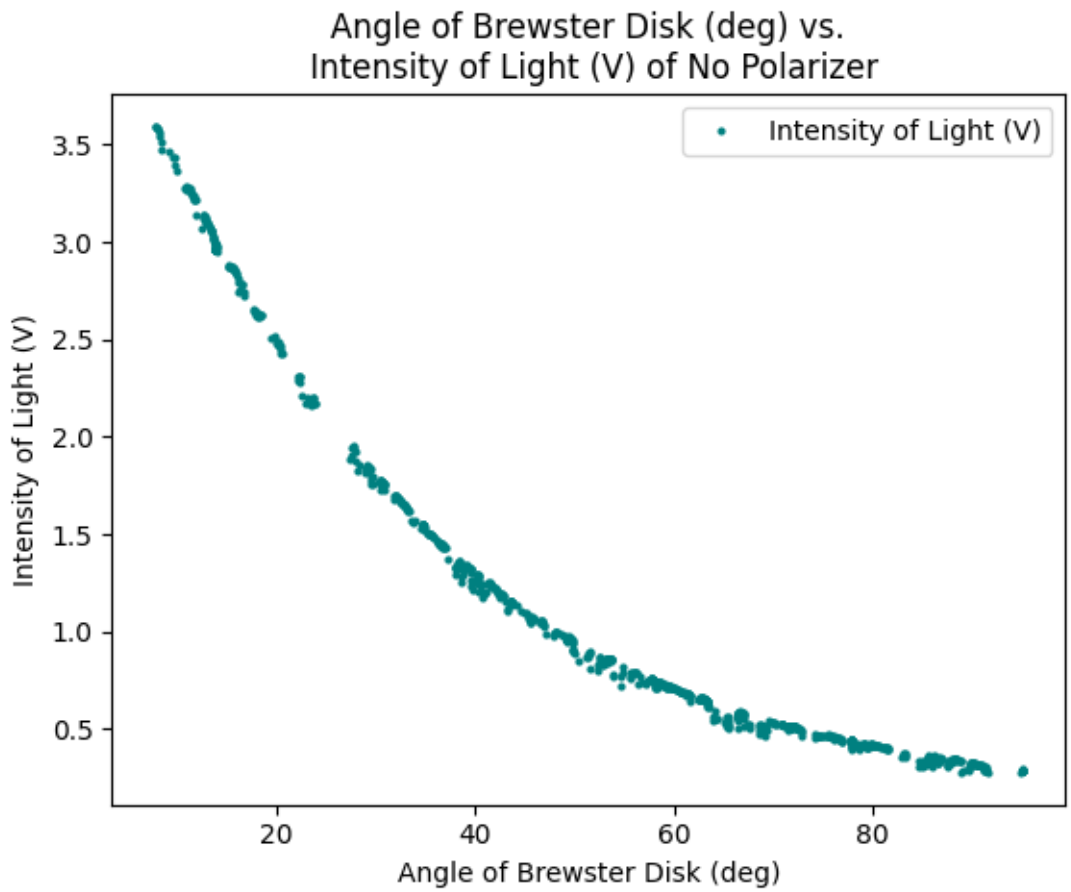


Brewster found at 68 deg. Uncertainty of 1 deg.

[19]:
```
plt.errorbar(x = vertical_position, y = vertical_intensity, fmt= 'o',
            ms=2, color='teal', ecolor='aquamarine', label="Intensity of Light␣
↪(V)")

# plt.plot(vertical_position, parallel_reflectance(vertical_position,␣
↪popt_v[0], popt_v[1], popt_v[2], popt_v[3]), label = "Triple Malus Law␣
↪$cos^2\\phi cos^2(a-\\phi)$ model", color='red')
```

```
# Legend and titles
plt.legend()
plt.xlabel("Angle of Brewster Disk (deg)")
plt.ylabel("Intensity of Light (V)")
plt.title("Angle of Brewster Disk (deg) vs.\n Intensity of Light (V) of␣
↪Vertical Polarizer")

plt.show()
```



[20]:
```
plt.errorbar(x = no_polarizer_position, y = no_polarizer_intensity, fmt= 'o',
             ms=2, color='teal', ecolor='aquamarine', label="Intensity of Light␣
↪(V)")

# plt.plot(no_polarizer_position, all_reflectance(no_polarizer_position,␣
↪popt_n[0], popt_n[1], popt_n[2], popt_n[3]), label = "Triple Malus Law␣
↪$cos^2\\phi cos^2(a-\\phi)$ model", color='red')
```

```
# Legend and titles
plt.legend()
plt.xlabel("Angle of Brewster Disk (deg)")
plt.ylabel("Intensity of Light (V)")
plt.title("Angle of Brewster Disk (deg) vs.\n Intensity of Light (V) of No␣
 ↪Polarizer")

plt.show()
```